Today's Content:-

→ Modulus Basics
→ Modulus Properties
→ Interesting Problems → Google

$\boxed{\begin{array}{lll} \text{int } x & \approx & [-2 * 10^9, 2 * 10^9] \\ \text{long } y & \approx & [-8 * 10^{18}, 8 * 10^{18}] \end{array}}$

$n \% a$ = remainder when n is divided by a

$10 \% 4 = 2$

$13 \% 5 = 3$

$\boxed{\text{Dividend} = \text{Divisor} * \text{Quotient} + \text{Remainder}}$

Remainder = Dividend − $\underbrace{(\text{Divisor} * \text{Quotient})}_{\text{Greatest multiple of divisor} <= \text{dividend}}$

Quiz 1

$150 \% 11 = 150 - \{\underbrace{\text{greatest multiple of } 11 <= 150}_{143}\}$

$= 150 - 143$

$= 7$

Quiz 2

$100 \% 7 = 100 - \{\text{greatest multiple of } 7 <= 100\}$

$= 100 - 98 = 2$

Quiz 3

$-40 \% 7 = -40 - \{\text{greatest multiple of } 7 <= -40\}$

-42

$= -40 - (-42)$

$= -40 + 42 = 2$

Quiz 5

$-60 \% 9 = -60 - \{\text{largest multiple of } 9 <= -60\}$

-63

$= -60 - (-63) = 3$

Python	C/C++/Java/JS
$-40 \% 7 = 2$	$-40 \% 7 = -5 + 7 = 2$
$-60 \% 9 = 3$	$-60 \% 9 = -6 + 9 = 3$
$-10 \% 3 = 2$	$-10 \% 3 = -1 + 3 = 2$

$a \% M$

if a is -ve, $a \% M = a \% M + M$

Why Modulus ?

It limits data to a required range.

$-\infty \ldots \infty \rightarrow \% 10 = \{0, 1, 2, 3, \ldots 8, 9, 10\}$

$[0, 9]$

$$\boxed{-\infty \ldots \infty \rightarrow \% M \rightarrow [0, M-1]}$$

→ Consistent Hashing.

→ Cryptography

Quiz

$(a+b) \% M \rightarrow [0, M-1]$

(1) $$\boxed{(a+b) \% M = ((a \% M + b \% M) \% M)}$$

$a=6, b=8, M=10$

$(6+8) \% 10$

$14 \% 10 = 4$

$6 \% 10 + 8 \% 10 = 6 + 8$

$= (14) \% 10$

$= 4$ → correct

$(a \% M + b \% M)$

$a \% M \rightarrow [0, M-1]$, $b \% M \rightarrow [0, M-1]$

$((a \% M + b \% M) \% M) \rightarrow [0, M-1]$

2) $\boxed{(a*b)\%M = (a\%M * b\%M)\%M}$

$a=6,\ b=8,\ M=10$

$(6*8)\%10$

$48\%10$

$=8$

$(6\%10)*(8\%10)$

$6*8 = (48)\%10$

$= 8$

3) $(a-b)\%M$

4) $(a/b)\%M$

$\Big\} \rightarrow$ Advance session

Ques1 You have to implement the power function (No in-built functions)

Given a, n, p. You have to calculate and return $a^n \% p$

Answers:-

~~(1<<a)%p~~

~~(a<<n)%p~~

~~(1<<n)%p~~

$a << n = a * 2^n \neq a^n$

$1 << n = 1 * 2^n \neq a^n$

$1 << a = 1 * 2^a \neq a^n$

```
power (a, n, p) {
    long ans = 1
    for(i=1 ; i<=N ; i++) {
        ans = ans * a;
    }
    return ans%p;
}
```

ans=1

i	ans
1	a
2	a^2
3	a^3
4	a^4
⋮	
N	a^N

Constraints:- $1 \le a \le 10^9$

$1 \le N \le 10^5$

$1 \le p \le 10^9$

e.g. $a=2;\ N=40,\ p=10^8$

$\rightarrow$ ans $= 2^{40}$ $\rightarrow$ can you store this?

e.g. $a=2;\ N=100;\ p=10^8$

$\rightarrow$ ans $= 2^{100}$ $\rightarrow$ long can't store this.

```
power (a, n, p) {
    long ans = 1
    for(i=1 ; i<=N ; i++) {
        ans = (ans * a)%p
    }
    return ans%p;
}
```

$\text{ans} = \boxed{\underbrace{(\text{ans} \% p)}_{[0, p-1]} * \underbrace{(a \% p)}_{[0, p-1]}} \% p$

max value $\approx p^2$

max $p = 10^9$

$(p^2)_{max} = 10^{18}$ $\rightarrow$ long can easily store this.

T.C = O(N)

S.C = O(1)

Dry run ⇒ a, N = 5, p $a^5 \% p$, ans = 1

i	ans = (ans * a) % p	ans
1	ans = (1 * a) % p	a % p
2	ans = (a % p * a) % p	a^2 % p
3	ans = (a^2 % p * a) % p	a^3 % p
4	ans = (a^3 % p * a) % p	a^4 % p
5	ans = (a^4 % p * a) % p	a^5 % p

Ques

① (a % p) % p = a % p

[0, p-1] % p

② (a * b) % p = (a % p * b) % p

x y

((a % p) % p * b % p) % p

(a % p * b % p) % p

— X — Break — X —

10:30

// Divisibility of 3

(sum of all digits) % 3 == 0

$1 \% 3 = 1$
$10 \% 3 = 1$
$10^2 \% 3 = 1$
$10^x \% 3 = 1$

$(3458) = (3*10^3 + 4*10^2 + 5*10^1 + 8*10^0)$

$(3458) \% 3 = (3*10^3 + 4*10^2 + 5*10^1 + 8*10^0) \% 3$

↓ Taking % inside

$((3*\underbrace{10^3) \% 3}_{1} + (4*\underbrace{10^2) \% 3}_{1} + (5*\underbrace{10^1) \% 3}_{1} + (8*\underbrace{10^0) \% 3}_{1}) \% 3$

$(3 + 4 + 5 + 8) \% 3$

if $(3+4+5+8) \% 3 == 0$
then $(3458) \% 3 == 0$

// Divisibility Rule for 9 = sum of digits % 9 == 0

$1 \% 9 = 1$
$10 \% 9 = 1$
$10^2 \% 9 = 1$
$10^3 \% 9 = 1$
$10^n \% 9 = 1$

Divisibility of 4

(last 2 digits) % 4 == 0

$1 \% 4 \neq 0$

$10 \% 4 \neq 0$

$100 \% 4 = 0$

$10^3 \% 4 = 0$

$10^x \% 4 = 0$

for all $x >= 2$

$4328 = (4 * 10^3 + 3 * 10^2 + 28)$

$4328 \% 4 = (4 * 10^3 + 3 * 10^2 + 28) \% 4$

$((4 * \cancel{10^3}) \% 4 + (3 * \cancel{10^2}) \% 4 + (28) \% 4) \% 4$

$\Rightarrow (28 \% 4) \% 4 \Rightarrow 28 \% 4$

Q Given 1 number in arr[], calculate number % p → GOOGLE

↳ each arr[i] contains single digit of the number.

Constraints:- $1 \le N \le 10^5$ → Array size

$1 \le p \le 10^9$

N=5

arr[5] =

0	1	2	3	4
3	2	4	6	4

⇒ 32464 % 4 = 0

p=4

Idea1:- Convert entire array to a number and then to that number you apply % p

$N=9 \rightarrow 9$ digits

$\approx 10^9 \rightarrow$ int

if $N=18 \rightarrow 18$ digits $\approx 10^{18} \rightarrow$ long

$N=100 \rightarrow 100$ digits $\approx 10^{100} \rightarrow$ ✗

$N=10^5 \rightarrow 10^5$ digits $\approx 10^{100000} \rightarrow$ ✗

Idea2:- p = 2, 3, 4, 5, 6, 7, 8, 9, 10 →

Impossible to do this using divisibility rules.

0	1	2	3	4	5
3	2	8	4	2	9

arr[6] =

$$(3*10^5 + 2*10^4 + 8*10^3 + 4*10^2 + 2*10^1 + 9*10^0)\ \%\ p$$

$$(3+2+8+4+2+9)\ \%\ p$$

→ only if $p=3$ or $p=9$

General, arr[N]

0	1	2	-	-	-	-	-	N-2	N-1

$$((arr[0]*10^{N-1})\ \%\ p + (arr[1]*10^{N-2})\ \%\ p + \ldots\ )\ \%\ p$$

$$(a*b)\ \%\ p$$

$$(a*(b\ \%\ p))\ \%\ p$$

$$(arr[0]*\boxed{(10^{N-1}\ \%\ p)})\ \%\ p$$

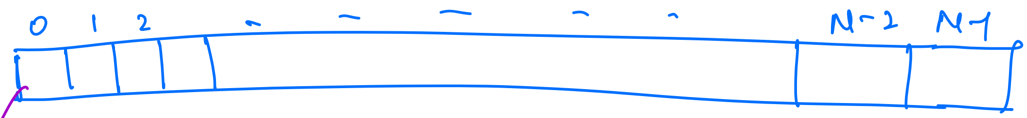
Math.pow(10, N-1) % p → X

10^{10^5} → 100% overflow.

→ power(a, n, p)

→ power(10, N-1, p)

→ (arr[0] * power(10, N-1, p)) % p



$$((arr[0] * 10^{N-1}) \% p + (arr[1] * 10^{N-2}) \% p + \cdots + (arr[N-1] * 10^{0}) \% p) \% p$$

$$(arr[0] * ((10^{N-1}) \% p)) \% p + (arr[1] * ((10^{N-2} \% p))) \% p + \cdots$$

$$(arr[0] * power(10, N-1, p)) \% p + (arr[1] * power(10, N-2, p))) \% p$$

Pseudo Code:-

```
ans = 0;
for (i=0; i < N; i++){
|    ans = [ans + ( arr[i]%p  * power(10, N-i-1, p)) %p] %p
}
return ans;
```

$T.C = O(N)$

Tarun

$T.C = O(N^2)$
$S.C = O(1)$

$N_{max} = 10^5$

$N^2 \rightarrow 10^{10} \longrightarrow$ TLE

Optimisation.

idea:-

	0	1	2	3	4	5
arr[6] =						
exp =	$10^5\%p$	$10^4\%p$	$10^3\%p$	$10^2\%p$	$10^1\%p$	$10^0\%p$

exp = 1

exp = (exp * 10) % p = 10 % p

$$exp = (exp * 10)\%p = ((10*10)\%p = 10^2\%p$$

Right to left

(Carry forward exponent value).

// Pseudo code

```
ans = 0
exp = 1
for (i = N-1; i >= 0; i--) {
    ans = (ans + (arr[i] * exp) % p) % p
    exp = (exp * 10) % p;
}
return ans;
```

T.C = O(N)

S.C = O(1)